Debabrata Mandal

COMPUTATIONAL OPTICS · GENERATIVE PHOTOGRAPHY · VIDEO DIFFUSION

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I am a fourth-year Ph.D. candidate at **UNC Chapel Hill** (VCAIL Lab, advised by Prof. Praneeth Chakravarthula), working at the intersection of **computational optics** and **generative AI**. I explore ways to couple physical imaging — metalens cameras, immersive stereo displays, optics-aware learning — with deep generative models for image restoration, video synthesis, and continual model adaptation. I have past experience working as an AI optimization engineer at KLA-Tencor.

Education

University of North Carolina at Chapel Hill *Ph.D., Computer Science* — Aug 2025 – Jul 2028
Chapel Hill, NC

Advisor: Prof. Praneeth Chakravarthula

University of North Carolina at Chapel Hill *M.S., Computer Science* — Jul 2023 – May 2025
Chapel Hill, NC

Indian Institute of Technology Bombay *B.Tech., Computer Science & Engineering* — Jun 2017 – Jun 2021
Mumbai, India

GPA: 4.34/5 | Collaborated with Prof. Parag Chaudhuri (ViGIL Lab)

Selected Publications

underline = first author

EPIC: Epipolar-Consistent 360° Immersive Stereo Video Generation — Under Review
D. Mandal, D. Fu, J. Miller, W. Villareal, X. Peng, P. Chakravarthula — arXiv:2609.38689

COLORS: Continual Learning with Consolidated Orthogonal LoRA Subspaces — Under Review
D. Mandal, Z. Peng, P. Chakravarthula — (double-blind)

DeltaCam: Differential Intrinsic Camera Modeling for Video Generation — arXiv 2026
D. Mandal, Z. Peng, Y. Wang, P. Chakravarthula — arXiv:2605.25266

UnSCAR: Universal, Scalable, Controllable, Adaptable Image Restoration — arXiv 2026
D. Mandal, S. Chattopadhyay, Y. Wang, M. Niethammer, P. Chakravarthula — arXiv:2603.07406

UniCoRN: Unified Controllable Image Restoration Network *(Oral)* — WACV 2026
D. Mandal, S. Chattopadhyay, G. Tong, P. Chakravarthula

High-Quality HDR Imaging from Extreme Degraded Metalens Bursts — WACV 2026
D. Mandal, Z. Peng, Y. Wang, P. Chakravarthula

Conditional Imaging for Privacy-Preserving Visual Sensing — arXiv 2026
H. Lian, D. Mandal, A. Wen, Y. Wang, X. Peng, P. Chakravarthula — arXiv:2610.28835

Broadband MWIR Imaging Enabled by All-Silicon Hyperboloid Metalens — arXiv 2026
R. Johnson, D. Mandal, Z. Zou, J. Wei, A. Wirth-Singh, V. Yarmolik, J. Froech, A. Majumdar, P. Chakravarthula — arXiv:2610.29841

Split-Knit Convolution: Dense Evaluation of Transpose Convolutions on GPUs — HiPC 2022
A. Menon Vadakkeveedu, D. Mandal, P. Ramachandran, N. Chandrachoodan

Work Experience

Dolby Laboratories — **Sight Experiences Lab** *PhD Research Intern* — Jun 2026 – Aug 2026
San Jose, CA

Investigated novel techniques for physically realistic stereoscopic video generation; developed EPIC, a 360° stereo pipeline extending video diffusion models to 4K immersive output via a pose-free epipolar consistency metric and DPO.

AMD — **Advanced Graphics Team** *Research Intern* — Jun 2025 – Aug 2025
Santa Clara, CA

Accelerated interactive generative rendering pipelines; optimised real-time GPU graphics performance.

KLA-Tencor — **Advanced Computing Labs** *AI Engineer* — Jun 2021 – Jul 2023
Chennai, India

Designed ML inference infrastructure for semiconductor inspection; GPU memory algorithms (w/ Dr. Pradeep Ramachandran & Prof. N. Chandrachoodan).

UNC Chapel Hill — VCAIL Lab *Graduate Research Assistant* — Aug 2023 – Present
Chapel Hill, NC

Research in computational imaging and computer vision; exploring deep-optics co-design, metalens imaging systems, and generative models for image reconstruction and restoration.

Awards & Service

Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) — 2026
LiFT'22 & LiFT'21 Scholar — Linux Foundation Advanced Open Source Training — 2021, 2022
Kaggle ML Research Spotlight — 2021
LFX Mentee, Open Horizon — Linux Foundation Edge — 2021
Bertelsmann Technology Scholarship — 2020